Semester: 6th
Subject Name:- A.I..& Code:- CS30002
Branch (s): - CSE

School of Computer Engineering
KIIT, Deemed to be University
Artificial Intelligence [CS30002]

SPRING MID SEMESTER EXAMINATION-2026

Total Time: 90 minutes **Full Marks:20**

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1. Answer all the questions [1 Mark × 5]

   a) Try to mention a scenario where breadth first search algorithm will give infeasible solution.
   b) How do ridges affect hill climbing performance? Why a plateau is problematic?
   c) When a Uniform Cost Search preferred over BFS and why?
   d) Why is simulated annealing considered similar to hill climbing when the temperature approaches zero?
   e) The heuristic path algorithm is a best-first search in which the objective function is $f(n) = (2 - w)g(n) + wh(n)$. For what values of w is this algorithm guaranteed to be optimal ? (You may assume that h is admissible.) What kind of search does this perform when $w = 0$, $w = 1$ and $w = 2$?

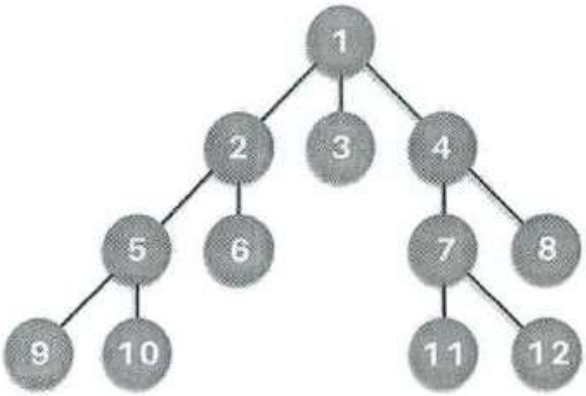
2. [3 + 2 Marks]

   a) An agent is placed in a partially observable and dynamic environment. Explain which type of intelligent agent would be most suitable and why. Justify your answer by discussing memory, decision-making and adaptability.

   b) Discuss why utility-based agents are considered more rational than goal-based agents, even when both achieve the same goal.

3. [5 Marks]

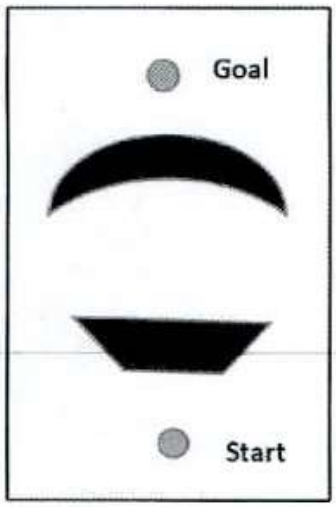
   Consider the following Graph:



   Apply (i) Breadth First Search(ii) Depth First Search and (iii) IDDFS (Iterative Deeping Depth-First Search) on the graph and compare them with respect to number of node explored. Consider 1 as start node and 12 as goal node.

4. [3 + 2 Marks]

a) A* Algorithm provides optimal path when the heuristics is admissible. Consider the below figure and show how an agent with partial observability can find a path from start to goal (circles) avoiding the obstacles (curved and trapezoid) using A* Algorithm. Consider that the agent can move up, down, right and left. You may create a grid to simplify the space or you may consider a continuous space.



b) Prove that every consistent heuristic is also admissible.

5. [3 + 2 Marks]

a) Provide the problem formulation for 8-Puzzle problem. Also demonstrate how Greedy Best First search algorithm will be applicable to this problem to find a solution.
b) Prove the given statement: "Depth-First Search is a special case of Best-First Search".

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